

THE WATERMARK IN EVERY EVEN DIMENSION

The discriminant of the Hodge lattice of a Fermat variety of prime degree

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A sequel to *The Watermark and Double Ladder Theorems* [WDL] and to *The Chaise Longue Theorem*, version 10 [CL]. Version 1 of this note stated the main formula as a conjecture and was not released; in this version it is a theorem.

Summary

For the Fermat surface of prime degree p , Shioda asked in 1987 whether the determinant of the Néron-Severi lattice is $p^{3(p-3)^2}$ [Shi87]. It is (the *Watermark* theorem [WDL], verified in Lean). This note gives the answer **in every even dimension**.

Let $X \subset P^{2k+1}$ be the Fermat variety of degree m and dimension $2k$, let $G = \mu_m^{2k+2}/\mu_m$ be its group of diagonal automorphisms, and let $Hdg(X)$ be the lattice of integral Hodge classes of middle degree.

1. **The discriminant group is a finite ring (Theorem 1, every degree $m \geq 3$).** $Hdg(X)^\vee/Hdg(X) \cong Z[G]/(I_{\mathfrak{g}} + I_T)$, where $I_{\mathfrak{g}}$ and I_T are the ideals of the elements of the group ring that vanish at the Hodge characters and at the remaining primitive characters. In words: the discriminant group is the coordinate ring of the intersection, inside $Spec Z[G]$, of the Hodge locus with its complement.
2. **The exponent is a delta invariant (Theorem 2, $m = p$ prime).** $|disc\ Hdg(X)| = p^E$ with $E = 1 + (2k(p-2) - 1) \cdot B - 2\delta_{\mathfrak{g}}$. Here B is the number of Hodge characters up to scalars and $\delta_{\mathfrak{g}}$ is the delta invariant of the order $Z_p[G]/I_{\mathfrak{g}}$, an arithmetic curve with B branches through one point.
3. **The formula (Theorem 3).** The tangent cone of that curve is the cone over the Hodge points $[\mathfrak{g}] \subset P^{2k}(F_p)$, and its coordinate ring modulo x_0^{p-1} is the ring of Theorem A of [CL] with the even box $p-1$. Hence $E_k(p) = 1 + (2k(p-2) - 1) \cdot B - 2 \cdot \sum_i [i/(p-1)] \cdot a(i)$, with a the Hilbert function of $F_p[x_0, \dots, x_{2k+1}]/(e_1, e_3, \dots, e_{2k+1}; x_i^{p-1})$. This step uses one count, the dimension of that ring, which follows from a theorem of Bezrukavnikov, Riche and Rider [BRR] as explained in [CL, Note]. Without it, the right-hand side is still a **lower bound** for $E_k(p)$; and in the 22 cells where the count was checked by machine the formula does not depend on [BRR].
4. **Closed forms.** For the Fermat cubic of dimension $2k$: $|disc\ Hdg(X)| = 3^{2 \cdot 4^k + 1 - 3 \cdot C(2k+1, k)}$. For the lattice spanned by the p^{k+1} linear spaces of one matching: discriminant $p^1 + (p-1)^k \cdot (k(p-2) - 1)$ (Theorem 4, with no use of [BRR]). For surfaces the formula returns $3(p-3)^2$.
5. **Verification.** The formula agrees with every published computation for prime degree known to us: ten cells (*dimension, degree*). For nine of them the tables of Aljovin, Movasati and Villaflor [AMV] give the lattice or its primitive part; (4, 7) is new, and so is the lattice of linear cycles at (6, 3). (For surfaces, the determinant had also been checked by computer for odd degrees up to 81 [SSvL].) Each identity used in the proof was checked by a separate engine that computes the orders directly in the group ring; that engine also confirms two cells for which no lattice computation exists, (6, 5) and (12, 3). Theorem

1 and its variant for the linear cycles (Remark (ii) of §2) were tested at composite degrees too: five cells against [AMV], with all the elementary divisors, and the Fermat sextic surface against [ABB].

What the proof rests on. Pham's theorem on the homology of the affine Fermat variety, in the form given by Looijenga [Loo, Corollary 2.2]; the Hodge type of the eigenspaces [SK] and Shioda's description of the Hodge characters for prime degree [Shi79], [Ran]; for Theorem 4, [DS, Corollary 1.6]; and, for Theorem 3 in general, [BRR, Proposition 2.12].

What is open. Two conjectures: that $E_k(p)$ is a polynomial in p of degree $k + 1$ (for $k \leq 7$ we give it; it agrees with every value computed, at primes up to 43); and that the transcendental side, too, is governed by its tangent cone, so that $E_k(p)$ is the length of the intersection of two cones over points of $P^{2k}(F_p)$. And three problems: the elementary divisors of the discriminant group, composite degrees, a proof without [BRR] (§9).

Status. Nothing here is refereed. The proofs were read cold by an independent reader before release; its findings are incorporated (§10).

1. Setting and notation

Throughout, $k \geq 1$, $n = 2k$, $N = 2k + 2$, $m \geq 3$, and $X = X_{2k}^m \subset P^{2k+1}$ is the Fermat variety $z_0^m + \dots + z_{N-1}^m = 0$. From §4 on, $m = p$ is an odd prime.

1.1 Lattices. $H := H_n(X, \mathbb{Z})$ with the intersection form is unimodular. $\eta \in H$ is the class of a linear section of dimension k ; $\eta \cdot \eta = m$. $H^\circ := \eta^\perp$ is the primitive homology. $\text{Hdg}(X) \subset H$ is the lattice of integral Hodge classes; it is primitive, it contains η , and the form is non-degenerate on it. $T(X) := \text{Hdg}(X)^\perp$ is the transcendental lattice; $T(X) \subset H^\circ$. Put $\text{Hdg}^\circ := \text{Hdg}(X) \cap H^\circ$. $L(X) \subset \text{Hdg}(X)$ is the sublattice generated by the classes of the $(2k+1)!! \cdot m^{k+1}$ standard linear k -spaces of X ; a standard space P has $P \cdot \eta = 1$.

- For m odd, $L(X)$ is primitive in H (Conjecture 1.2 of Degtyarev and Shimada [DS], proved in [CL, Main Theorem]).
- For $m = p$ prime, $\text{Hdg}(X) = L(X)$, of rank $Q_k(p) + 1$ [CL, Corollary H]. This fact is not used below; only the description of the Hodge characters in 1.2 is. Here $Q_k(p)$ is the number of vectors of $(F_p \setminus 0)^N$ whose coordinates can be matched in pairs $\{a, -a\}$; it is the number of closed walks of length N on $\mathbb{Z}^{(p-1)/2}$ with steps $\pm \varepsilon_i$.

1.2 Characters. $G := \mu_m^N / \mu_m$ acts on X . Let $u_0, \dots, u_{N-1} \in G$ be the images of the generators of the factors; $u_0 \cdot u_1 \cdots u_{N-1} = 1$. The characters of G are the vectors $a = (a_0, \dots, a_{N-1}) \in (\mathbb{Z}/m)^N$ with $\sum a_\nu = 0$; $\chi_a(u_\nu) = \zeta^{a_\nu}$, $\zeta = e^{2\pi i/m}$. Write

- $\mathcal{A}$ for the characters with no zero coordinate (the *primitive* characters);
- $\mathcal{B} \subset \mathcal{A}$ for the *Hodge* characters: those a with $\sum_\nu \{t \cdot a_\nu / m\} = k + 1$ for every unit t of $\mathbb{Z}/m$ [SK], [Shi79];
- $T := \mathcal{A} \setminus \mathcal{B}$.

For $m = p$ prime, $\mathcal{B}$ is the set of the $a \in \mathcal{A}$ whose coordinates can be matched in pairs $\{a_i, a_j\}$ with $a_i + a_j = 0$ [Shi79], [Ran]; $|\mathcal{B}| = Q_k(p)$.

For a set Y of characters, $I_Y := \{x \in \mathbb{Z}[G] : \chi_a(x) = 0 \text{ for every } a \in Y\}$, an ideal of $\mathbb{Z}[G]$.

1.3 Pham's theorem, after Looijenga. For a G -lattice M with an invariant form, put $a \star b := \sum_{g \in G} (a \cdot gb) \cdot g \in \mathbb{Z}[G]$, so that $a \cdot b$ is the coefficient of 1 in $a \star b$.

Theorem (Pham [Ph]; Looijenga [Loo, Proposition 2.1 and Corollary 2.2]). H° is a cyclic $\mathbb{Z}[G]$ -module, generated by the image e of the Pham cycle, and the intersection form is

$(-1)^{n(n+1)/2} \star$ with

$$e \star e = \theta := \prod_{v=0}^{N-1} (1 - u_v) \in Z[G].$$

So for $x, y \in Z[G]$: $(xe) \cdot (ye) = \pm \varepsilon(x \cdot \bar{y} \cdot \theta)$, where ε is the coefficient of 1 and $\bar{y}$ is the image of y under $g \mapsto g^{-1}$. Since $\chi_a(\theta) = \prod_v (1 - \zeta^{a_v})$ is non-zero exactly for $a \in \mathfrak{A}$, and the form is non-degenerate on H° , the annihilator of e is $I_{\mathfrak{A}}$:

$$H^\circ \cong Z[G]/I_{\mathfrak{A}}, x \cdot e \mapsto x.$$

Under this isomorphism, $H^\circ \otimes C$ is the sum of the eigenlines of the characters in $\mathfrak{A}$, and the eigenline of a has Hodge type (k, k) exactly when $\Sigma_v \{a_v/m\} = k + 1$ [SK]. We claim

$$\text{Hdg}^\circ = I_T \cdot e \text{ and } T(X) = I_{\mathfrak{B}} \cdot e. \quad (1)$$

Proof. A class $x \cdot e \in H^\circ$ is rational, so the set of the characters a with $\chi_a(x) \neq 0$ is stable under $a \mapsto ta$, $t \in (Z/m)^*$. Hence $x \cdot e$ is a Hodge class if and only if that set consists of characters all of whose multiples ta have type (k, k) , that is, lies in $\mathfrak{B}$; that is, $x \in I_T$ (modulo $I_{\mathfrak{A}}$). For the second equality: $y \cdot e$ is orthogonal to $I_T \cdot e$ if and only if $\varepsilon(x \cdot \bar{y} \cdot \theta) = 0$ for every $x \in I_T$; since I_T is an ideal, this means $x \cdot \bar{y} \cdot \theta = 0$ for every $x \in I_T$, that is, $\chi_a(x) \cdot \chi_{-a}(y) \cdot \chi_a(\theta) = 0$ for every a . For $a \in \mathfrak{B}$ there is an $x \in I_T$ with $\chi_a(x) \neq 0$ (for instance $|G|$ times the sum of the idempotents of the multiples ta of a), and $\chi_a(\theta) \neq 0$; so the condition is $\chi_a(y) = 0$ on $-\mathfrak{B} = \mathfrak{B}$. ■

(Both $\mathfrak{B}$ and T are stable under $a \mapsto -a$ and under $a \mapsto ta$, $t \in (Z/m)^*$, so (1) does not depend on the conventions that identify characters on homology and on cohomology.) For surfaces, (1) is [ABB, Proposition 2.10], where Looijenga's description is used to compute the transcendental lattice of the Fermat sextic.

2. The discriminant group is a finite ring

Theorem 1. Let $m \geq 3$ and $n = 2k \geq 2$. As $Z[G]$ -modules,

$$\text{Hdg}(X)^\vee / \text{Hdg}(X) \cong Z[G]/(I_{\mathfrak{B}} + I_T).$$

In particular $|\text{disc } \text{Hdg}(X)| = |\text{disc } T(X)| = |Z[G]/(I_{\mathfrak{B}} + I_T)|$.

Proof. (a) Since H is unimodular and $\text{Hdg}(X)$ is primitive and non-degenerate, the map $H \rightarrow \text{Hdg}(X)^\vee$, $x \mapsto (x \cdot -)$, is surjective with kernel $T(X)$. So $\text{Hdg}(X)^\vee / \text{Hdg}(X) \cong H / (\text{Hdg}(X) \oplus T(X))$, and the same holds for $T(X)$.

(b) The inclusion $H^\circ \subset H$ induces an injection $H^\circ / (\text{Hdg}^\circ \oplus T(X)) \rightarrow H / (\text{Hdg}(X) \oplus T(X))$, because $H^\circ \cap (\text{Hdg}(X) \oplus T(X)) = \text{Hdg}^\circ \oplus T(X)$. It is surjective: a standard space P lies in $\text{Hdg}(X)$ and has $P \cdot \eta = 1$, so every $x \in H$ is $(x \cdot \eta) \cdot P$ plus an element of H° ; hence $H = \text{Hdg}(X) + H^\circ$.

(c) By (1), $H^\circ / (\text{Hdg}^\circ \oplus T(X)) = Z[G]e / (I_T e + I_{\mathfrak{B}} e) \cong Z[G]/(I_T + I_{\mathfrak{B}})$, since $I_{\mathfrak{A}} \subset I_T \cap I_{\mathfrak{B}}$. All the maps are G -equivariant. ■

Remarks. (i) The ring $\mathcal{D} := Z[G]/(I_{\mathfrak{B}} + I_T)$ is finite, because $\mathfrak{B}$ and T are disjoint. Its spectrum is the scheme-theoretic intersection of the closed subschemes $V(I_{\mathfrak{B}})$ and $V(I_T)$ of $\text{Spec } Z[G]$: of the Hodge characters and of the others. It is supported at the primes dividing m .

(ii) The same proof gives, for every $m \geq 3$: if $\mathfrak{D} \subset \mathfrak{A}$ is the set of the characters whose coordinates can be matched in pairs $\{a_i, a_j\}$ with $a_i + a_j = 0$ (for even m this allows $a_i = a_j = m/2$), then $\Lambda^\vee/\Lambda \cong Z[G]/(I_{\mathfrak{D}} + I_{\mathfrak{A} \setminus \mathfrak{D}})$ for the saturation Λ of $L(X)$ in H . What replaces (1) is that $\Lambda \cap H^\circ = I_{\mathfrak{A} \setminus \mathfrak{D}} \cdot e$: the m^{k+1} standard spaces of a matching J form one orbit of G , with stabilizer $\{g : g_i = g_j \text{ on the pairs of } J\}$, so the characters that occur in $L(X) \otimes C$ are trivial on one of these stabilizers, that is, they are 0 or lie in $\mathfrak{D}$; and the rank of $L(X)$ is $|\mathfrak{D}| + 1$ [DS, Theorem 1.4]. For m odd, $\Lambda = L(X)$ by [CL, Main Theorem]; for surfaces of degree $m \leq 4$ or prime to 6 this is due to Degtyarev [Deg], and [DS, §5] verified it by computer for $(n, m) = (4, m)$ with $3 \leq m \leq 12$ and for $(6, 3), (6, 4), (6, 5), (8, 3)$.

(iii) We have not found Theorem 1 in the literature (nor did the cold reader of §10, who searched for it). The description (1) of the two lattices inside the group ring is known ([Loo]; [ABB, Proposition 2.10] for surfaces), and the proof of Theorem 1 is a few lines from there, so it may well be known to the specialists; if a reference exists, it should replace this paragraph.

Check. The group $Z[G]/(I_{\mathfrak{g}} + I_T)$ was computed in ten cells (§8, engine 6, Table 4): it is $(Z/8)^2$ for the Fermat quartic surface, $(Z/5)^{10} \times Z/25$ at $(2, 5)$, $Z/3 \times Z/9$ at $(4, 3)$, $(Z/3)^7 \times (Z/9)^7 \times Z/27$ at $(6, 3)$. These are the discriminant groups of $Hdg(X)$ in [AMV, Table 1] and in Table 1 below. For the Fermat sextic surface, where the lines do not generate the Néron–Severi lattice, it is $Z/4 \times (Z/12)^9 \times (Z/36)^9 \times Z/108$, of order $2^{40} \cdot 3^{30}$: the discriminant of the transcendental lattice found in [ABB, Remark 3.5]. Remark (ii) was tested in the same way at $(2, 6), (2, 8), (2, 9), (2, 10)$ and $(4, 4)$: in each cell the group is the one of [AMV, Table 1], with all its elementary divisors.

3. Orders, delta invariants and discriminants

From here on $m = p$ is an odd prime, $\zeta = e^{2\pi i/p}$, $\pi := \zeta - 1$, and $r := N - 1 = 2k + 1$, so $|G| = p^r$.

3.1 The orders. Let $Y \subset \mathfrak{A}$ be stable under multiplication by F_p^* , and $[Y]$ its set of orbits (a set of points of the projective space of characters, $P^{2k}(F_p)$). Choose a representative a in each orbit and put

$$\tilde{O}_Y := \prod_{[a] \in [Y]} Z[\zeta], A_Y := \text{the image of } Z[G] \text{ in } \tilde{O}_Y \text{ under } x \mapsto (\chi_a(x))_a.$$

So $A_Y \cong Z[G]/I_Y$. The index of A_Y in $\tilde{O}_Y$ is finite and a power of p (after inverting p , the group ring of the p -group G is a product of rings $Z[1/p][\zeta]$ and $Z[1/p]$). Define

$$\delta_Y := \log_p [\tilde{O}_Y : A_Y].$$

$\text{Spec } A_Y \otimes Z_p$ is a curve with $|[Y]|$ branches, each a copy of $\text{Spec } Z_p[\zeta]$, through one closed point, and δ_Y is its delta invariant. Write $\alpha := |[Y]|$, $B := |[Y]| = Q_k(p)/(p-1)$, $t := |[T]| = \alpha - B$.

Lemma 3.2 (intersection number). If $Y = Y_1 \sqcup Y_2$, then $\log_p |Z[G]/(I_{Y_1} + I_{Y_2})| = \delta_Y - \delta_{Y_1} - \delta_{Y_2}$.

Proof. $A_Y \subset A_{Y_1} \times A_{Y_2} \subset \tilde{O}_{Y_1} \times \tilde{O}_{Y_2} = \tilde{O}_Y$, and $(x, y) \mapsto x - y$ induces an isomorphism $(A_{Y_1} \times A_{Y_2})/A_Y \cong Z[G]/(I_{Y_1} + I_{Y_2})$. ■

Lemma 3.3 (discriminant of a lattice supported on Y). Let $M \subset H^\circ$ be a sublattice of the form $M = m \cdot e$, with $m \subset Z[G]$ a subgroup whose elements vanish at every character of $\mathfrak{A} \setminus Y$, and M of rank $|Y|$. Let $M_Y \subset \tilde{O}_Y \otimes Q$ be the image of m . Then

$$v_p(\text{disc } M) = -\kappa \cdot |[Y]| + 2 \cdot \log_p [\tilde{O}_Y : M_Y], \text{ with } \kappa := 2k(p-2) - 1.$$

(For lattices M_Y not contained in $\tilde{O}_Y$ the index is the generalized index.)

Proof. For $x, y \in m$, $(xe) \cdot (ye) = \pm \varepsilon(xy\bar{\theta}) = \pm p^{-r} \cdot \sum_{a \in Y} \chi_a(x) \cdot \chi_a(\bar{y}) \cdot \chi_a(\theta) = \pm p^{-r} \cdot \text{Tr}(x \cdot \bar{y} \cdot \theta)$, where Tr is the trace of the Q -algebra $K_Y := \tilde{O}_Y \otimes Q$ and $\bar{y}$ is complex conjugation on each factor. So $\text{disc } M = \pm p^{-r|Y|} \cdot N_{K_Y/Q}(\theta) \cdot \text{disc}_{\text{Tr}}(M_Y)$ (the conjugation changes the determinant by a sign). Now $|Y| = (p-1) \cdot |[Y]|$; $\text{disc}_{\text{Tr}}(M_Y) = \text{disc}(\tilde{O}_Y) \cdot [\tilde{O}_Y : M_Y]^2$ and $\text{disc } Z[\zeta] = \pm p^{p-2}$; and $N_{Q(\zeta)/Q}(1 - \zeta) = p$ gives $N_{K_Y/Q}(\theta) = \pm p^{N \cdot |[Y]|}$. The exponent of p is $[-r(p-1) + (p-2) + N] \cdot |[Y]| + 2 \cdot \log_p [\tilde{O}_Y : M_Y]$, and $r(p-1) - (p-2) - N = 2k(p-2) - 1$. ■

Theorem 2. Let p be an odd prime, $k \geq 1$, and $E := v_p(\text{disc } \text{Hdg}(X^p_{2k}))$. Then $|\text{disc } \text{Hdg}(X)| = p^E$ and

- (i) $2\delta_{\mathcal{Y}} = 1 + \kappa \cdot \alpha$;
- (ii) $E = 1 + \kappa \cdot B - 2\delta_{\mathcal{Y}}$;
- (iii) $E = \kappa \cdot t - 2\delta_T$;
- (iv) $E = \delta_{\mathcal{Y}} - \delta_{\mathcal{Y}} - \delta_T$.

Here $\kappa = 2k(p-2) - 1$.

Proof. $|\text{disc } \text{Hdg}(X)|$ is the order of $Z[G]/(I_{\mathcal{Y}} + I_T)$ by Theorem 1, a power of p ; and (iv) is Theorem 1 with Lemma 3.2.

(i) $|\text{disc } H^\circ| = p$: $Z\eta \oplus H^\circ$ has index p in the unimodular lattice H , and $\eta \cdot \eta = p$. Apply Lemma 3.3 to $M = H^\circ$, $m = Z[G]$, $Y = \mathcal{Y}$, $M_Y = A_{\mathcal{Y}}$.

(ii) $Z\eta \oplus \text{Hdg}^\circ$ has index p in $\text{Hdg}(X)$ (the kernel of $x \mapsto x \cdot \eta \bmod p$ on $\text{Hdg}(X)$), so $|\text{disc } \text{Hdg}^\circ| = p^{E+1}$. By (1), $\text{Hdg}^\circ = I_T \cdot e$, and its image in $\tilde{O}_{\mathcal{Y}}$ is the ideal $I_T \cdot A_{\mathcal{Y}}$ of $A_{\mathcal{Y}}$, of index $|Z[G]/(I_{\mathcal{Y}} + I_T)| = p^E$. So $[\tilde{O}_{\mathcal{Y}} : M_{\mathcal{Y}}] = p^{\delta_{\mathcal{Y}} + E}$, and Lemma 3.3 gives $E + 1 = -\kappa B + 2\delta_{\mathcal{Y}} + 2E$.

(iii) The same with $T(X) = I_{\mathcal{Y}} \cdot e$ and $|\text{disc } T(X)| = p^E$: $E = -\kappa t + 2\delta_T + 2E$. ■

Remark. The equality of orders in Theorem 1 has, for prime degree, a second proof. Put $D := \log_p |Z[G]/(I_{\mathcal{Y}} + I_T)|$. Without Theorem 1, Lemma 3.3 gives $E + 1 = -\kappa B + 2\delta_{\mathcal{Y}} + 2D$ and $E = -\kappa t + 2\delta_T + 2D$ (the second uses only $|\text{disc } T(X)| = |\text{disc } \text{Hdg}(X)|$, step (a) of the proof of Theorem 1). Adding them, with (i) and Lemma 3.2, gives $E = D$. Statement (iv) says that E is the intersection number at p of the arithmetic curve of the Hodge characters with the curve of the others, in the form « δ of the union minus the δ 's of the parts».

The same for any set of characters. Let $Y \subset \mathcal{Y}$ be stable under F_p^* , and $\Lambda_Y \subset H$ the lattice of the classes whose components lie in the eigenspaces of Y and of the trivial character. If Λ_Y contains a class of degree 1, the proofs of Theorem 1 and of (ii) apply word for word, with Λ_Y , $\Lambda_Y \cap H^\circ = I_{\mathcal{Y} \setminus Y} \cdot e$ and $\Lambda_Y^\perp = I_Y \cdot e$ in the place of $\text{Hdg}(X)$, Hdg° and $T(X)$: $|\text{disc } \Lambda_Y| = |Z[G]/(I_Y + I_{\mathcal{Y} \setminus Y})|$ is a power of p , and

$$v_p(\text{disc } \Lambda_Y) = 1 + \kappa \cdot |[Y]| - 2\delta_Y. \quad (2)$$

4. The tangent cone of the Hodge order

It remains to compute $\delta_{\mathcal{Y}}$. Let $S := F_p[x_1, \dots, x_r]$ be the ring of polynomial functions on the space of characters, and $x_0 := -(x_1 + \dots + x_r)$. For Y as in 3.1 let $I([Y]) \subset S$ be the homogeneous ideal of the points $[Y]$, $S_Y :=$

$S/I([Y])$, and $h_{[Y]}$ its Hilbert function; $h_{[Y]}(j) = |[Y]|$ for j large.

Theorem 3. (a) $\delta_{\mathfrak{g}} \leq \sum_{j \geq 0} (B - h_{[\mathfrak{g}]}(j))$. Hence

$$E_k(p) \geq 1 + \kappa \cdot B - 2 \cdot \sum_{j \geq 0} (B - h_{[\mathfrak{g}]}(j)) \geq 1 + \kappa \cdot B - 2 \cdot \sum_i [i/(p-1)] \cdot a(i),$$

where a is the Hilbert function of the ring in (C) below.

(b) Suppose that

$$(C) \dim_{F_p} F_p[x_0, \dots, x_{N-1}]/(e_1, e_3, \dots, e_{N-1}; x_0^{p-1}, \dots, x_{N-1}^{p-1}) = Q_k(p),$$

where e_j is the j -th elementary symmetric polynomial. Then equality holds in (a), and

$$E_k(p) = 1 + (2k(p-2) - 1) \cdot B - 2 \cdot \sum_i [i/(p-1)] \cdot a(i) = 1 - B + 2 \cdot \sum_{j=0}^{k(p-2)-1} h(j),$$

where a is the Hilbert function of the ring in (C) and $h(j) := \sum_{s \geq 0} a(j - s(p-1))$.

(c) (C) holds for every odd prime p and every k by [CL, Note, Proposition 3.1], which rests on [BRR, Proposition 2.12]. Independently of [BRR], (C) was verified by machine in the 22 cells of Table 6.

The ring in (C) is the ring of Theorem A of [CL] with the even box $p-1$. By [CL, Note] it is, over an infinite field, the space of coinvariants of the centralizer (as an algebraic group) of a regular unipotent element of Sp_{p-1} in the $(2k+2)$ -nd tensor power of the vector representation.

4.1 Proof of (a). Tensor everything with Z_p ; from here on $Z[G]$, $I_{\mathfrak{g}}$, $A_{\mathfrak{g}}$, $\tilde{O}_{\mathfrak{g}}$ stand for their tensor products with Z_p . Filter $\tilde{O}_{\mathfrak{g}}$ by the powers of π : $gr \tilde{O}_{\mathfrak{g}} = \Pi_{[a]} F_p[\varpi]$, with ϖ the class of π . Give $A_{\mathfrak{g}} \subset \tilde{O}_{\mathfrak{g}}$ the induced filtration $A_{\mathfrak{g}} \cap \pi^j \tilde{O}_{\mathfrak{g}}$. Since $A_{\mathfrak{g}} \supset \pi^J \tilde{O}_{\mathfrak{g}}$ for J large,

$$\delta_{\mathfrak{g}} = \sum_{j \geq 0} (B - \dim gr_j A_{\mathfrak{g}}). \quad (3)$$

Put $y_{\nu} := u_{\nu} - 1 \in Z[G]$. Its image in $\tilde{O}_{\mathfrak{g}}$ is $(\zeta^{a_{\nu}} - 1)_a$, and $\zeta^{a_{\nu}} - 1 \equiv a_{\nu} \cdot \pi \pmod{\pi^2}$. So for a homogeneous $f \in S$ of degree j , the element $f(y_1, \dots, y_r)$ (with coefficients lifted to Z) lies in $\pi^j \tilde{O}_{\mathfrak{g}}$, and its class in gr_j is $(f(a) \cdot \varpi^j)_a$. The map $f \mapsto (f(a) \varpi^{\deg f})_a$ has kernel $I([\mathfrak{B}])$. Therefore $S_{\mathfrak{g}} \subset gr A_{\mathfrak{g}}$, $\dim gr_j A_{\mathfrak{g}} \geq h_{[\mathfrak{g}]}(j)$, and the first inequality of (a) follows from (3) and Theorem 2(ii).

For the second inequality put $z := x_0^{p-1}$. $S_{\mathfrak{g}}$ embeds in $\Pi_{[a]} F_p[\varpi]$, where z acts as ϖ^{p-1} ; so $S_{\mathfrak{g}}$ is a finitely generated graded torsion-free module over $F_p[z]$, hence free, on generators whose degrees are counted by the Hilbert function a' of $S_{\mathfrak{g}}/zS_{\mathfrak{g}}$. So $h_{[\mathfrak{g}]}(j) = \sum_{s \geq 0} a'(j - s(p-1))$, $\sum_{i \equiv j \pmod{p-1}} a'(i) = B$ for every j , and $\sum_j (B - h_{[\mathfrak{g}]}(j)) = \sum_i [i/(p-1)] \cdot a'(i)$. The forms $e_j(x_0, \dots, x_{N-1})$ with j odd and $x_{\nu}^{p-1} - x_0^{p-1}$ vanish on $[\mathfrak{B}]$ (proof of Lemma 4.4), so the ring in (C) maps onto $S_{\mathfrak{g}}/zS_{\mathfrak{g}}$, and $a' \leq a$ in every degree. ■

4.2 Three lemmas for (b). On $Z_p[G]$ let Fil^j be the Z_p -span of the elements $p^c \cdot y^e$ (y^e a monomial in $y_1, \dots, y_r$) with $c(p-1) + |e| \geq j$. It is a decreasing filtration by ideals, $Fil^i \cdot Fil^j \subset Fil^{i+j}$, and Fil^j maps into $\pi^j \tilde{O}_{\mathfrak{g}}$. Note that $y_0 \in Fil^1$ and $y_0 \equiv -(y_1 + \dots + y_r) \pmod{Fil^2}$, because $u_0 = \Pi_{i \geq 1} (1 + y_i)^{-1}$. For a form $f \in S$ of degree d we write $f(y)$ for $f(y_1, \dots, y_r)$ with the coefficients lifted to Z_p ; another lift changes it by p times a form of degree d , an element of Fil^{d+p-1} .

Lemma 4.3 (lifts). Put $N_{\nu} := 1 + u_{\nu} + \dots + u_{\nu}^{p-1}$ and $\tau_{\nu} := u_{\nu} - u_{\nu}^{-1}$. Then:

- $N_{\nu} \in I_{\mathfrak{g}} \subset I_{\mathfrak{g}}$, and $N_{\nu} \equiv p + y_{\nu}^{p-1} \pmod{Fil^p}$;
- for j odd, $E_j := 2^{-j} \cdot e_j(\tau_0, \dots, \tau_{N-1}) \in I_{\mathfrak{g}}$, and $E_j \equiv e_j(y_0, \dots, y_{N-1}) \pmod{Fil^{j+1}}$.

Proof. $\chi_a(N_{\nu}) = 0$ when $a_{\nu} \neq 0$, and $N_{\nu} = ((1 + y_{\nu})^p - 1)/y_{\nu} = y_{\nu}^{p-1} + p + \sum_{i=2}^{p-1} C(p, i) \cdot y_{\nu}^{i-1}$, where each term of the sum has weight at least p . For $a \in \mathfrak{B}$ the numbers $s_{\nu} := \zeta^{a_{\nu}} - \zeta^{-a_{\nu}}$ come in pairs $\{s, -s\}$, so $\Pi_{\nu} (1 + s_{\nu} X)$ is

even in X and the odd elementary symmetric functions of the s_ν vanish: $\chi_a(E_j) = 0$. Finally $\tau_\nu = (1 + y_\nu) - (1 + y_\nu)^{-1} \equiv 2y_\nu \pmod{\text{Fil}^2}$. ■

Lemma 4.4 (the ideal of the Hodge points). Assume (C). Then $I([\mathcal{B}])$ is generated by the forms $e_j(x_0, \dots, x_{N-1})$, j odd, $3 \leq j \leq N-1$, and $x_\nu^{p-1} - x_0^{p-1}$, $1 \leq \nu \leq N-1$. Moreover $S_{\mathcal{B}}$ is a free module over $F_p[x_0^{p-1}]$, with a basis that lifts a homogeneous basis of the ring in (C); in particular $h_{[\mathcal{B}]}(j) = h(j)$.

Proof. Let $\mathcal{S}$ be the quotient of S by those forms. They vanish on $[\mathcal{B}]$: at a point with all coordinates in F_p^* , $x_\nu^{p-1} = x_0^{p-1}$; and on a character that can be matched in pairs the odd elementary symmetric functions vanish. So $\mathcal{S} \rightarrow S_{\mathcal{B}}$ is surjective. Put $z := x_0^{p-1}$. $S_{\mathcal{B}}$ embeds in $\prod_{[a]} F_p[\varpi]$, where z acts as ϖ^{p-1} , so $S_{\mathcal{B}}$ is a finitely generated graded torsion-free $F_p[z]$ -module, hence free; its rank is $(p-1) \cdot B = Q_k(p)$ (the cokernel of the embedding has finite length, and $\prod F_p[\varpi]$ has rank $(p-1)B$). On the other hand $\mathcal{S}/z\mathcal{S}$ is the ring in (C), of dimension $Q_k(p)$, so by the graded Nakayama lemma $\mathcal{S}$ is generated over $F_p[z]$ by $Q_k(p)$ homogeneous elements. A surjection $F_p[z]^Q \rightarrow \mathcal{S} \rightarrow S_{\mathcal{B}} \cong F_p[z]^Q$ is an isomorphism. ■

Lemma 4.5 (the induced filtration is the quotient filtration). Assume (C). Then $A_{\mathcal{B}} \cap \pi^j \tilde{O}_{\mathcal{B}}$ is the image of Fil^j , for every j .

Proof. Let $w \in Z_p[G]$ have image in $\pi^j \tilde{O}_{\mathcal{B}}$ and let $d \leq j$ be maximal with $w \in \text{Fil}^d + I_{\mathcal{B}}$. Suppose $d < j$, and write $w = w_d + w' + i$ with $i \in I_{\mathcal{B}}$, $w' \in \text{Fil}^{d+1}$ and $w_d = \sum c_{c,e} \cdot p^c y^e$ over the pairs with $c(p-1) + |e| = d$ (coefficients in Z_p). Since $p \equiv -\pi^{p-1} \pmod{\pi^p}$ in $Z[\zeta]$ (Wilson), the class of the image of w_d in $\text{gr}_d \tilde{O}_{\mathcal{B}}$ is that of the form $f := \sum \bar{c}_{c,e} \cdot (-x_0^{p-1})^c x^e \in S_d$; it is zero because $d < j$, so $f \in I([\mathcal{B}])_d$ (f may be the zero form, when the terms of w_d cancel; the argument covers this case). By Lemma 4.4, $f = \sum \varphi_i g_i$ with g_i among the generators listed there. By Lemma 4.3 each g_i has a lift $\tilde{g}_i \in I_{\mathcal{B}}$ with $\tilde{g}_i \equiv g_i(y) \pmod{\text{Fil}^{\deg g_i + 1}}$ (namely E_j and $N_\nu - N_0$). So $v := \sum \varphi_i(y) \cdot \tilde{g}_i \in I_{\mathcal{B}}$ and $v \equiv f(y) \pmod{\text{Fil}^{d+1}}$. Also $p^c y^e - (-y_0^{p-1})^c y^e \in N_0 \cdot \text{Fil}^{d-(p-1)} + \text{Fil}^{d+1} \subset I_{\mathcal{B}} + \text{Fil}^{d+1}$. Hence $w \in \text{Fil}^{d+1} + I_{\mathcal{B}}$, against the choice of d . ■

Remark (tangent cone). In $A_{\mathcal{B}}$ the relation $N_0 = 0$ reads $p \cdot (1 + c) = -y_0^{p-1}$ with c in the maximal ideal $\mathfrak{m}$ of $A_{\mathcal{B}}$; so $p \in \mathfrak{m}^{p-1}$, and the image of Fil^j in $A_{\mathcal{B}}$ is $\mathfrak{m}^j$. With Lemma 4.5, the filtration induced from $\tilde{O}_{\mathcal{B}}$ is the $\mathfrak{m}$ -adic one, and $\text{gr}_{\mathfrak{m}} A_{\mathcal{B}} = S_{\mathcal{B}}$; the cone over $[\mathcal{B}]$ is the tangent cone of $\text{Spec } A_{\mathcal{B}}$ at its closed point, in the usual sense.

4.6 Proof of (b). By Lemma 4.5, $\text{gr}_j A_{\mathcal{B}}$ is spanned by the classes of the $p^c y^e$ with $c(p-1) + |e| = j$, which lie in the image of S (the class of p is that of $-x_0^{p-1}$). With 4.1, $\text{gr } A_{\mathcal{B}} = S_{\mathcal{B}}$ and (3) gives $\delta_{\mathcal{B}} = \sum_j (B - h_{[\mathcal{B}]}(j))$. By Lemma 4.4, $h_{[\mathcal{B}]}(j) = \sum_{s \geq 0} a(j - s(p-1))$, and $\sum_{i \equiv j \pmod{p-1}} a(i) = B$ for each residue class. So $B - h(j) = \sum_{i > j, i \equiv j} a(i)$ and $\sum_j (B - h(j)) = \sum_i [i/(p-1)] \cdot a(i)$. This is the first form.

For the second form we need $a(i) = 0$ for $i > (k+1)(p-2)$. Over Q , multiplication by $e_1 = x_0 + \dots + x_{N-1}$ on $Q[x]/(x_i^{p-1})$ is the raising operator of an $\mathfrak{sl}_2$ -module in which a monomial of degree i has weight $2i - N(p-2)$; it is onto the part of positive weight, so the quotient by e_1 vanishes in degrees $> N(p-2)/2$. The Hilbert function of the ring in (C) is the same over F_p and over Q : each graded piece of the ring over Z has rank at most its dimension modulo p , and the total dimensions agree ((C), and [CL, Note, Proposition 3.1] in characteristic 0). Hence $h(j) = B$ for $j \geq k(p-2)$, and $\sum_j (B - h(j)) = k(p-2) \cdot B - \sum_{j < k(p-2)} h(j)$. ■

5. One matching, cubics, surfaces

Theorem 4 (one matching). Let p be an odd prime, J a matching of $\{0, \dots, N-1\}$ in pairs, and $\Lambda_J \subset H$ the lattice spanned by the classes of the p^{k+1} standard spaces of J . Then

$$|\text{disc } \Lambda_J| = p^{1 + (p-1)^k \cdot (k(p-2) - 1)}.$$

Proof. Let $Y = \mathfrak{B}_J$ be the set of the characters with $a_i + a_j = 0$ for the pairs of J and no zero coordinate; $|Y| = (p-1)^{k+1}$. The p^{k+1} spaces of J form one orbit of G , with stabilizer $\{g : g_i = g_j \text{ on the pairs of } J\}$; so the characters that occur in $\Lambda_J \otimes C$ are trivial on that subgroup, that is, they have $a_i + a_j = 0$ on the pairs, and those that occur in H are 0 and the elements of Y . Λ_J is primitive, of rank $(p-1)^{k+1} + 1$ [DS, Corollary 1.6]; so all of them occur and Λ_J is the lattice Λ_Y of (2). It contains a standard space, of degree 1. $[Y]$ is the set of the $(p-1)^k$ points of a projective space $P^k(F_p)$ with no zero coordinate; its ideal is generated by the linear forms $x_i + x_j$ and the forms $w_i^{p-1} - w_0^{p-1}$ in the coordinates $w_0, \dots, w_k$ of that space (a reduced complete intersection of the right degree). These generators lift to I_Y with the right leading forms: $u_i u_j - 1 \equiv y_i + y_j \text{ mod } \text{Fil}^2$, and $N_v - N_0$. So the proof of Lemma 4.5 applies without (C), $\text{gr } A_Y = S_Y$, and $\delta_Y = \Sigma_j ((p-1)^k - h_{[Y]}(j))$ with $\Sigma_j h_{[Y]}(j) X^j = (1 + X + \dots + X^{p-2})^k / (1 - X)$. The sum is the derivative at 1 of $(1 + X + \dots + X^{p-2})^k$, that is $\delta_Y = k(p-1)^k(p-2)/2$. By (2), $|\text{disc } \Lambda_J|$ is a power of p and $v_p(\text{disc } \Lambda_J) = 1 + (2k(p-2) - 1)(p-1)^k - k(p-1)^k(p-2)$. ■

Remark (a second computation of δ_Y , due to the cold reader). $A_Y = Z[G]/I_Y$ is the group ring of $(Z/p)^{k+1}$ modulo the cyclotomic polynomials $\Phi_p(t_i)$, that is, $Z[\zeta]^{\otimes(k+1)}$, whose trace form has discriminant $\text{disc}(Z[\zeta])^{(k+1)(p-1)^k}$, against $\text{disc}(Z[\zeta])^{(p-1)^k}$ for $\tilde{O}_Y$. Hence $2\delta_Y = k(p-2)(p-1)^k$, with no use of §4.

Corollary 5 (Fermat cubics). For every $k \geq 1$, $|\text{disc } \text{Hdg}(X^3_{2k})| = 3^{2 \cdot 4^k + 1 - 3 \cdot C(2k+1, k)}$: the exponents are 0, 3, 24, 135, 663, 3045, 13464,

Proof. For $p = 3$ the ring in (C) is $F_3[x]/(e_0 dd; x_i^2)$, and (C) holds by [CL, Note, Proposition 3.1] (here the group is SL_2); for $2 \leq k \leq 6$ it was also verified by machine. Its Hilbert function is that over Q . Modulo the squares, Newton's identities read $j \cdot e_j = e_1 \cdot e_{j-1}$, so over Q the ring is $Q[x]/(e_1; x_i^2)$, and by the sl_2 argument of 4.6, $a(i) = C(N, i) - C(N, i-1)$ for $i \leq k+1$ and $a(i) = 0$ beyond. Hence $h(j) = C(2k+1, j)$ for $j \leq k$, $B = C(2k+1, k)$, and $E = 1 - B + 2 \cdot \Sigma_{j=0}^{k-1} C(2k+1, j) = 1 - C(2k+1, k) + 2 \cdot (4^k - C(2k+1, k))$. ■

For $k = 2$ this is 27, the row $1^{19} \cdot 3 \cdot 9$ of [AMV, Table 1].

Surfaces. For $k = 1$, $[\mathfrak{B}]$ is the set of the $3(p-2)$ points with no zero coordinate on the three lines $x_0 + x_i = 0$ of $P^2(F_p)$: $p-1$ points on each line, the three vertices of the triangle counted once. A form of degree $j \leq p-2$ that vanishes on $[\mathfrak{B}]$ vanishes on the three lines, so $h_{[\mathfrak{B}]}(j) = 3j$ for $1 \leq j \leq p-2$, and $\Sigma_j (B - h_{[\mathfrak{B}]}(j)) = (3p^2 - 9p + 4)/2$. Theorem 3(b) then gives $E_1(p) = 1 + (2p-5) \cdot 3(p-2) - (3p^2 - 9p + 4) = 3(p-3)^2$: the Watermark theorem [WDL], which has an independent proof verified in Lean.

6. The numbers

The polynomials (observed, not proved). With $x = p - 3$:

k	dimension	E_k , as a polynomial in $x = p - 3$
1	2	$3x^2$
2	4	$30x^3 + 30x + 3$
3	6	$315x^4 - 420x^3 + 1225x^2 - 504x + 24$
4	8	$3780x^5 - 14175x^4 + 47250x^3 - 64890x^2 + 40830x + 135$
5	10	$51975x^6 - 374220x^5 + 1767150x^4 - 4546080x^3 + 6464271x^2 - 3788994x + 663$
6	12	$810810x^7 - 9459450x^6 + 63693630x^5 - 256351095x^4 + 628533906x^3 - 855737883x^2 + 492167858x + 3045$
7	14	$14189175x^8 - 243243000x^7 + 2260808550x^6 - 13038905880x^5 + 48478915485x^4 - 112810998300x^3 + 148705398465x^2 - 83774680984x + 13464$

For $k = 1$ this is a theorem for every p . For $k \geq 2$, each row was obtained by interpolation on $k + 2$ primes and then tested on at least four primes not used in the fit; no test failed (ajuste. log; the primes go up to 43).

What was interpolated. The values used are those of the walk statistic (engine 4 of §8). That statistic is the Hilbert function of the ring in (C) — hence, by Theorem 3, gives $E_k(p)$ — in the 22 cells of Table 6, where the ring itself was computed; for the even box this identification is proved nowhere in general ([CL, Note, Theorem 7.1] proves it for odd boxes). So the rows are the formula of Theorem 3 at the cells of Table 6 ($k = 2$: the ten primes 3, ..., 31; $k = 3$: 3, 5, 7, 11; $k = 4$: 3, 5; $k = 5$, 6: 3), the closed form of Corollary 5 at $p = 3$ for every k , and the walk statistic elsewhere. In every row the leading coefficient is $k \cdot (2k+1)!!$ and the constant term is $E_k(3)$. That E_k is a polynomial in p is Conjecture P of §9.

Values. In the following cells there was no lattice computation. By Theorem 3 (with (C) verified by machine in each of them, except the last, which is Corollary 5):

variety	disc Hdg(X)	rank
Fermat fourfold of degree 11	$\pm 11^{15603}$	10901
Fermat fourfold of degree 13	$\pm 13^{30303}$	19921
Fermat sixfold of degree 5	$\pm 5^{5596}$	4901
Fermat sixfold of degree 7	$\pm 7^{71368}$	44731
Fermat eightfold of degree 5	$\pm 5^{94395}$	63505
Fermat cubic of dimension 12	$\pm 3^{3045}$	3433
Fermat cubic of dimension 14	$\pm 3^{13464}$	12871

Two of them, the sixfold of degree 5 and the cubic of dimension 12, were confirmed by the direct computation of $\delta_{\mathcal{H}}$ in the group ring (Table 2), which uses Theorem 2 only.

A law that is false. Let t be the number of Galois orbits in T , the rank over $Z[\zeta_p]$ of the transcendental lattice. On every surface $E = 3(p-3)^2 = 3t$, and on the Fermat cubics $E_k(3) = 3t$ as well. But at $(4, 5)$, $E = 303$ and $3t = 315$; at $(4, 7)$, $E = 2043$ and $3t = 2403$. By Theorem 2(iii) the correct statement is $E = \kappa \cdot t - 2\delta_T$.

7. What the theorems say, in one paragraph

$\text{Spec } Z[G]$ is a union of arithmetic curves, one for each Galois orbit of characters, all meeting over p . The Hodge characters cut out one curve, the other primitive characters another. Theorem 1 says that the discriminant group of $\text{Hdg}(X)$ is the ring of functions on the intersection of these two curves. Theorem 2 computes its length by the classical formula for the intersection number of two curves through their delta invariants; the form $\theta = \Pi(1 - u_v)$ of Pham contributes the constant κ . Theorem 3 says that the Hodge curve is as singular as its tangent cone, the cone over the Hodge points in $P^{2k}(F_p)$; and that cone is the one object that [CL] is about, the union of the $(2k+1)!!$ matching subspaces, with the box $p - 1$. For a single matching the cone is a complete intersection and everything is explicit (Theorem 4).

8. Verification

All the runs that are used stay under 1.2 GB and 10 minutes on a laptop; the runs stopped by the guard are not used, and their logs are kept and say so. Scripts and logs: `engines/` (in the repository; `corpus4/regla296_conjetura/` and `corpus4/regla297_teorema/` in the working tree).

Seven engines.

1. **Lattice.** The Gram matrix of all standard k -spaces ($P \cdot P' = (1 - (1-p)^c)/p$, with $c - 1$ the dimension of $P \cap P'$), then its Smith form over Z/p^M . It reproduces the published rows of [AMV, Table 1] with the complete lists of elementary divisors.
2. **Points.** The Hilbert functions of the point sets $[A], [B], [T] \subset P^{2k}(F_p)$, by linear algebra over F_p .
3. **Ring.** The Hilbert function of the ring in (C) by Macaulay2, over F_p (and over Q and F_{32003} in some cells).
4. **Walks.** The same Hilbert function from the fibre-rank statistic on closed walks of [CL, Note]; no Gröbner basis. It agrees with engine 3 in the 22 cells where both ran.
5. **Orders.** New in this version. The matrix of the evaluations $Z[G] \rightarrow \tilde{O}_Y$ for $Y = A, B, T$, and its Smith form over Z/p^M : this gives $\delta_A, \delta_B, \delta_T$ straight from the definition, with no geometry and no Hilbert function.
6. **Discriminant group.** New. The abelian group $Z[G]/(I_B + I_T)$, for any degree m , by two programs: exactly over Z (integer kernels and a Smith form; small cells), and prime by prime from the indices of Lemma 3.2 (`grupo_lector_en_frio.py`, written by the cold reader of §10 and run again here).
7. **Pham model.** New. The Gram matrices of $H^\circ = Z[G]e$, of $\text{Hdg}^\circ = I_T \cdot e$ and of $T(X) = I_B \cdot e$, built from the form θ of §1.3 and nothing else, and their Smith forms at p . This tests §1.3 from end to end: it has to return $|\text{disc } H^\circ| = p$ and the published rows of [AMV, Table 2].

Table 1. The exponent $E = v_p(\text{disc Hdg}(X))$: lattice against Theorem 3.

(n, p)	elementary divisors of $\text{Hdg}(X) = L(X)$ (p -part)	E , lattice	source of the lattice value	E , Theorem 3
(2, 5)	$1^{26} \cdot 5^{10} \cdot 25$	12	[AMV, Table 1]; engine 1	12
(2, 7)	$1^{48} \cdot 7^{38} \cdot 49^5$	48	[AMV, Table 1]; engine 1	48
(2, 11)	$1^{96} \cdot 11^{158} \cdot 121^{17}$	192	[AMV, Table 1]	192
(2, 13)	$1^{120} \cdot 13^{254} \cdot 169^{23}$	300	[AMV, Table 1]	300
(4, 3)	$1^{19} \cdot 3 \cdot 9$	3	[AMV, Table 1]; engine 1	3
(4, 5)	$1^{166} \cdot 5^{174} \cdot 25^{54} \cdot 125^7$	303	[AMV, Table 1]; engine 1	303
(4, 7)	$1^{443} \cdot 7^{854} \cdot 49^{504} \cdot 343^{59} \cdot 2401$	2043	engine 1, new (two runs, independent pivot orders)	2043
(6, 3)	$1^{56} \cdot 3^7 \cdot 9^7 \cdot 27$	24	engine 1, new ; [AMV, Table 2] gives $\text{Hdg}^\circ: 1^{54} \cdot 3^8 \cdot 9^7 \cdot 27$, exponent $25 = 24 + 1$	24
(8, 3)	$\text{Hdg}^\circ: 1^{172} \cdot 3^{35} \cdot 9^{34} \cdot 27^{11}$	135	[AMV, Table 2], $136 - 1$	135
(10, 3)	$\text{Hdg}^\circ: 1^{559} \cdot 3^{144} \cdot 9^{144} \cdot 27^{76} \cdot 81$	663	[AMV, Table 2], $664 - 1$	663

In the last three rows we use $|\text{disc Hdg}^\circ| = p \cdot |\text{disc Hdg}(X)|$ (proof of Theorem 2(ii)). Engine 1 computes the lattice $L(X)$ of the linear spaces; $L(X) = \text{Hdg}(X)$ in these cells by [Shi79] and the primitivity of $L(X)$, which [DS, §5] verified by computer at (4, 3), (4, 5), (4, 7), (6, 3) (and [CL] proves in general).

Table 2. Engine 5: the delta invariants, and the four identities of Theorem 2.

(n, p)	α	B	t	κ	$\delta_{\mathfrak{A}}$	$\delta_{\mathfrak{B}}$	δ_T	$2\delta_{\mathfrak{A}} - 1 - \kappa\alpha$	$1 + \kappa B - 2\delta_{\mathfrak{B}}$	$\kappa t - 2\delta_T$	$\delta_{\mathfrak{A}} - \delta_{\mathfrak{B}} - \delta_T$
(2, 5)	13	9	4	5	33	17	4	0	12	12	12
(2, 7)	31	15	16	9	140	44	48	0	48	48	48
(2, 11)	91	27	64	17	774	134	448	0	192	192	192
(2, 13)	133	33	100	21	1397	197	900	0	300	300	300
(4, 3)	11	10	1	3	17	14	0	0	3	3	3
(4, 5)	205	100	105	11	1128	399	426	0	303	303	303
(6, 3)	43	35	8	5	108	76	8	0	24	24	24
(8, 3)	171	126	45	7	599	374	90	0	135	135	135
(10, 3)	683	462	221	9	3074	1748	663	0	663	663	663
(4, 7)	1111	310	801	19		1924			2043		
(6, 5)	3277	1225	2052	17		7615			5596		
(12, 3)	2731	1716	1015	11		7916			3045		

In the last three rows only $\delta_{\mathfrak{B}}$ was computed. For (6, 5) and (12, 3) no lattice computation exists; the values of $\delta_{\mathfrak{B}}$ are those that Theorem 3 gives from the Hilbert function of the ring.

Table 3. Theorem 4 against the lattice (engine 1 restricted to one matching).

(n, p)	rank	elementary divisors (p-part)	$v_p(\text{disc}), \text{lattice}$	$1 + (p-1)^k(k(p-2) - 1)$
(2, 7)	37	$1^{12} \cdot 7^{25}$	25	25
(4, 5)	65	$1^{11} \cdot 5^{27} \cdot 25^{27}$	81	81
(4, 7)	217	$1^{17} \cdot 7^{75} \cdot 49^{125}$	325	325
(4, 11)	1001	$1^{29} \cdot 11^{243} \cdot 121^{729}$	1701	1701
(6, 3)	17	$1^6 \cdot 3^6 \cdot 9^4 \cdot 27$	17	17
(6, 5)	257	$1^{14} \cdot 5^{54} \cdot 25^{108} \cdot 125^{81}$	513	513

Table 4. Theorem 1: the group $Z[G]/(I_{\mathfrak{g}} + I_T)$ (engine 6) against the discriminant group of $\text{Hdg}(X)$.

(n, m)	$Z[G]/(I_{\mathfrak{g}} + I_T)$	discriminant group of $\text{Hdg}(X)$	source
(2, 4)	$(Z/8)^2$	$(Z/8)^2$	[AMV, Table 1], row $1^{18} \cdot 8^2$
(2, 5)	$(Z/5)^{10} \times Z/25$	$(Z/5)^{10} \times Z/25$	[AMV, Table 1]
(4, 3)	$Z/3 \times Z/9$	$Z/3 \times Z/9$	[AMV, Table 1]
(6, 3)	$(Z/3)^7 \times (Z/9)^7 \times Z/27$	$(Z/3)^7 \times (Z/9)^7 \times Z/27$	engine 1
(2, 6)	$Z/4 \times (Z/12)^9 \times (Z/36)^9 \times Z/108$, of order $2^{40} \cdot 3^{30}$	order $2^{40} \cdot 3^{30}$	[ABB, Remark 3.5] (the discriminant of $T(X)$; the group is not given there)

Table 4 bis. Remark (ii) to Theorem 1 at composite degrees: the group $Z[G]/(I_{\mathfrak{g}} + I_{\mathfrak{g} \setminus \mathfrak{g}})$ (matchable characters) against the lattice of linear cycles of [AMV, Table 1].

(n, m)	$Z[G]/(I_{\mathfrak{g}} + I_{\mathfrak{g} \setminus \mathfrak{g}})$	[AMV, Table 1]
(2, 6)	$(Z/4)^{12} \times (Z/3)^{21} \times (Z/9)^3 \times Z/27$	$3^{13} \cdot 12^8 \cdot 36^3 \cdot 108$
(2, 8)	$(Z/2)^{12} \times (Z/8)^{48} \times (Z/16)^2 \times (Z/32)^8 \times (Z/64)^4$	$2^{12} \cdot 8^{48} \cdot 16^2 \cdot 32^8 \cdot 64^4$
(2, 9)	$(Z/3)^7 \times (Z/9)^{72} \times (Z/27)^7 \times (Z/81)^{11}$	$3^7 \cdot 9^{72} \cdot 27^7 \cdot 81^{11}$
(2, 10)	$(Z/2)^{96} \times (Z/4)^{24} \times (Z/5)^{127} \times (Z/25)^{10} \times Z/125$	$5^{18} \cdot 10^{96} \cdot 20^{13} \cdot 100^{10} \cdot 500$
(4, 4)	$(Z/2)^2 \times (Z/4)^4 \times (Z/8)^{30} \times (Z/16)^4 \times (Z/32)^2$	$2^2 \cdot 4^4 \cdot 8^{30} \cdot 16^4 \cdot 32^2$

The two columns are the same groups. At (4, 4) the Hodge characters and the matchable characters give the same group.

Table 5. Engine 7: the lattices of §1.3 built from Looijenga's form (p-parts of the elementary divisors).

(n, p)	H°	$\text{Hdg}^\circ = I_T \cdot e$	$T(X) = I_{\mathfrak{g}} \cdot e$	against
(2, 5)	$1^{51} \cdot 5$	$1^{24} \cdot 5^{11} \cdot 25$	$1^5 \cdot 5^{10} \cdot 25$	$T(X)$: the group of $\text{Hdg}(X)$ in [AMV, Table 1]
(4, 3)	$1^{21} \cdot 3$	$1^{17} \cdot 3^2 \cdot 9$	$3 \cdot 9$	the same
(6, 3)	$1^{85} \cdot 3$	$1^{54} \cdot 3^8 \cdot 9^7 \cdot 27$	$1 \cdot 3^7 \cdot 9^7 \cdot 27$	Hdg° : the row (6, 3) of [AMV, Table 2]; $T(X)$: engine 1
(8, 3)	$1^{341} \cdot 3$	$1^{172} \cdot 3^{35} \cdot 9^{34} \cdot 27^{11}$	$1^{11} \cdot 3^{34} \cdot 9^{34} \cdot 27^{11}$	Hdg° : the row (8, 3) of [AMV, Table 2]

So the discriminant group of $Hdg(X)$ for the Fermat cubic eightfold is $(\mathbb{Z}/3)^{34} \times (\mathbb{Z}/9)^{34} \times (\mathbb{Z}/27)^{11}$, of order 3^{135} ; [AMV] give the primitive lattice only.

Table 6. Condition (C), by machine (engine 3, in characteristic p). $\dim = Q_k(p)$ and top degree $(k+1)(p-2)$ in the cells (n, p) : (2, 5), (2, 7), (2, 11), (2, 13); (4, 3), (4, 5), (4, 7), (4, 11), (4, 13), (4, 17), (4, 19), (4, 23), (4, 29), (4, 31); (6, 3), (6, 5), (6, 7), (6, 11); (8, 3), (8, 5); (10, 3); (12, 3).

9. Two conjectures and three problems

Conjecture P (the Watermark polynomials). For every $k \geq 1$ there is a polynomial $W_k \in \mathbb{Z}[x]$ of degree $k + 1$, with leading coefficient $k \cdot (2k+1)!!$ and constant term $2 \cdot 4^k + 1 - 3 \cdot C(2k+1, k)$, such that $E_k(p) = W_k(p - 3)$ for every odd prime p . For $k \leq 7$, W_k is the polynomial in the table of §6.

For $k = 1$ this is the Watermark theorem. For $2 \leq k \leq 7$ the evidence is the table of §6: W_k was fitted on $k + 2$ primes and agrees with the values at the other primes computed ($k = 2$: fitted on 29, 31, 37, 41, tested on 3, 5, 7, 11, 13, 17, 19, 23; $k = 3$: tested on 3, ..., 19; $k = 4$: on 3, ..., 17; $k = 5, 6, 7$: on 3, 5, 7, 11). As explained in §6, outside the cells of Table 6 these values are those of the walk statistic; that the statistic computes the Hilbert function of the ring with an even box is part of the conjecture. By Theorem 3 the conjecture is a statement about the Hilbert functions of the rings in (C), that is, about the generalized exponents of the representations of sp_{p-1} that occur in the $(2k+2)$ -nd tensor power of the vector representation ([CL, Note]; [Lus]).

Conjecture S (the two curves meet as their tangent cones do). For $Y = T$ and for $Y = \mathcal{Y}$, the delta invariant of the order A_Y is that of its tangent cone: $\delta_Y = \sum_{j \geq 0} (|f_Y| - h_{f_Y}(j))$.

For $Y = \mathcal{Y}$ this is Theorem 3(b). By Theorem 2 the left-hand sides are known: $\delta_T = (\kappa t - E)/2$ and $\delta_{\mathcal{Y}} = (1 + \kappa\alpha)/2$; so Conjecture S is a statement about the Hilbert functions of two explicit sets of points of $P^{2k}(F_p)$. It holds in the thirteen cells where those Hilbert functions were computed: (2, 5), (2, 7), (2, 11), (2, 13), (4, 3), (4, 5), (4, 7), (4, 11), (6, 3), (6, 5), (8, 3), (10, 3), (12, 3) (conos_tangentes_A_T.log). With Theorem 2(iv) and Theorem 3 it is equivalent to the pair of statements $\sum_j (\alpha - h_{f_{\mathcal{Y}}}(j)) = (1 + \kappa\alpha)/2$ and

$$E_k(p) = \dim_{F_p} S/(I([\mathcal{Y}]) + I([T])) = \sum_j [h_{f_{\mathcal{Y}}}(j) + h_{f_T}(j) - h_{f_{\mathcal{Y}}}(j)]:$$

the exponent of the discriminant is the length of the intersection of the cone over the Hodge points with the cone over the others. In those cells the algebra $S/(I([\mathcal{Y}]) + I([T]))$ has a symmetric Hilbert function, with last non-zero degree $2k(p-2) - 2$; for surfaces it has the Hilbert function of a complete intersection of type $(3, p-3, p-3)$, whose Bézout number is $3(p-3)^2$.

Problem 1 (the discriminant group). Theorem 1 gives the discriminant group as a ring, not its elementary divisors. For surfaces they are given by the *Double Ladder* theorem [WDL]. The tables show the first rows in higher dimension; for instance the exponent of the group (the characteristic of the ring $\mathcal{D}$) is p^2 for surfaces, p^3 at (4, 5) and p^4 at (4, 7). In Table 3 the groups follow a visible pattern in $u = p - 2$ (in dimension 4: $(\mathbb{Z}/p)^{3u^2} \times (\mathbb{Z}/p^2)^{u^3}$), recorded here and not claimed.

Problem 2 (other degrees). Theorem 1 holds for every m . For m a prime power the ring $\mathbb{Z}_p[G]$ is still local and the method of §3–§4 should apply, with the constant κ replaced by a sum over the orbits; for general

composite m several primes interact. For composite m the lattice $Hdg(X)$ can be larger than the saturation of $L(X)$: at $(2, 6)$ the ranks are 86 and 62 ([ABB]; [AMV, Table 1]). None of this is done here.

Problem 3. A proof of (C) that does not use [BRR] ([CL, v10, Problem 7] is its odd-box companion).

10. Status, and what is not claimed

- **Proved, modulo [Ph]/[Loo, Corollary 2.2] and the classical facts on Fermat varieties quoted in §1** ([SK] for the Hodge type of the eigenlines; [Shi79], [Ran] for the Hodge characters of prime degree; [DS, Corollary 1.6] for Theorem 4): Theorem 1 (every $m \geq 3$), Theorem 2, Theorem 3(a), Theorem 4.
 - **Proved, modulo [Ph]/[Loo] and [BRR, Proposition 2.12]:** Theorem 3(b) for every k and every odd prime p ; Corollary 5.
 - **Proved, modulo [Ph]/[Loo] and a machine computation of one dimension:** Theorem 3(b) in the 22 cells of Table 6; in particular the values of §6.
 - **Not proved:** the polynomials of §6 for $k \geq 2$ (they interpolate the walk statistic at primes up to 43, which is the formula of Theorem 3 in the cells of Table 6: Conjecture P); Conjecture S; the problems of §9.
 - **The sign** of the discriminant is not addressed.
 - **Read in the original (arXiv versions):** [Loo], Proposition 2.1 and Corollary 2.2; [ABB], Proposition 2.10 and Remark 3.5; [AMV], Tables 1 and 2; [DS]; [BRR], §2 (for [CL, Note]). **Not read in the original:** [Ph] (quoted through [Loo]); [SK], [Shi79], [Ran], [Lus], [Kos] (quoted as in [CL] and [DS]); [Deg] (quoted as in [WDL]).
 - **Unrefereed inputs:** [BRR] is an arXiv preprint; [CL], [CL, Note] and [WDL] are the author's and are not refereed.
 - **Cold reading.** A first complete draft of this version was read cold by an independent reader (a separate AI session, without access to the working files, with the mandate to break it and to write its own code). Its verdict: *holds with corrections*; no fatal error. It re-derived Theorems 1–4, Lemmas 3.2–4.5, §4.6 and Corollary 5 line by line; read [Loo] and [BRR, Proposition 2.12] (statement, hypotheses and proof, not its inputs) and checked that the hypotheses hold for Sp_{p-1} in characteristic p ; recomputed with its own engines Tables 1–4, except $\delta_{\mathcal{B}}$ at $(6, 5)$, and condition (C) in sixteen cells; and tested Theorem 1 at composite degrees (Table 4 bis is its addition). It found four errors of statement (the sentence on Hodge types in §1.3, with the second half of (1) not argued; a false sentence on $Hdg(X) \neq L(X)$; «Theorem 1 is proved twice»; the description of what the polynomials interpolate), three small gaps (Theorem 4, Remark (ii), the lower bound in terms of a) and thirteen points of presentation. All of them are corrected in this text. Added after the reading, and so not read cold: engine 7 and Table 5; the rewriting of the open problems as Conjectures P and S, with the separate check of the two tangent cones. The reader could not verify: the inputs of [BRR]; (C) outside the cells computed; the polynomials for $k = 6, 7$; the originals of [SK], [Shi79], [Ran]. Its report, scripts and logs are in the repository.
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11. Reproducibility

- `gram_smith.py n p [M]` — engine 1 ($SUBJ=0$ for one matching, $PERM=seed$ for a second pivot order).
- `hilbert2.py n p full|short` — engine 2.

- `ta_even.py` `k p [char]` — engine 3 (needs Macaulay2).
- `E_estadistica2.py` `k, p ...` — engine 4; `ajuste.py` — the interpolation and its tests.
- `delta_ordenes.py` `n p [M]` — engine 5 (*RED=1* uses only the group elements with coordinates in $0..p-2$, which span each order over Z_p ; *SETS=B* computes δ_g only).
- `grupo_discriminante.py` `n p`, `grupo_general.py` `n m`, `grupo_lector_en_frio.py` `n m H|D` — engine 6.
- `pham_gram.py` `n p` — engine 7.

Longest runs: $(6, 5)$, δ_g , 388 s and 1.0 GB; $(4, 7)$ lattice, 105 s and 0.5 GB. Runs stopped by the guard are not used; their logs end with the line of the guard that says so.

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References

- **[ABB]** A. Auel, C. Böhning, H.-C. Graf von Bothmer, *The transcendental lattice of the sextic Fermat surface*, Math. Res. Lett. **20** (2013), no. 6, 1017–1031. arXiv:1306.6798. (Proposition 2.10 and Remark 3.5, read in the arXiv version.)
- **[AMV]** E. Aljovin, H. Movasati, R. Villafior Loyola, *Integral Hodge conjecture for Fermat varieties*, J. Symbolic Comput. **95** (2019), 177–184. doi:10.1016/j.jsc.2019.02.006; arXiv:1711.02628. (Tables 1 and 2 were read in the original.)
- **[BRR]** R. Bezrukavnikov, S. Riche, L. Rider, *Modular affine Hecke category and regular unipotent centralizer*, arXiv:2005.05583 (v2, 4 July 2024). (Proposition 2.12.)
- **[CL]** R. Amichis Luengo, *The Chaise Longue Theorem*, version 10, Zenodo (2026). doi:10.5281/zenodo.22961150. With its supplementary note *The Chaise Longue Theorem and the centralizer of a regular unipotent element*, cited as [CL, Note].
- **[Deg]** A. Degtyarev, *Lines generate the Picard groups of certain Fermat surfaces*, J. Number Theory **147** (2015), 454–477. doi:10.1016/j.jnt.2014.07.020; arXiv:1305.3073. (Quoted as in [WDL].)
- **[DS]** A. Degtyarev, I. Shimada, *On the topology of projective subspaces in complex Fermat varieties*, J. Math. Soc. Japan **68** (2016), no. 3, 975–996. doi:10.2969/jmsj/06830975; arXiv:1405.4683.
- **[Kos]** B. Kostant, *Lie group representations on polynomial rings*, Amer. J. Math. **85** (1963), 327–404.
- **[Loo]** E. Looijenga, *Fermat varieties and the periods of some hypersurfaces*, in: Algebraic and Arithmetic Structures of Moduli Spaces (Sapporo 2007), Adv. Stud. Pure Math. **58** (2010), 47 ff. doi:10.2969/aspm/05810047; arXiv:1005.1733. (Proposition 2.1 and Corollary 2.2.)
- **[Lus]** G. Lusztig, *Singularities, character formulas, and a q-analog of weight multiplicities*, Astérisque **101–102** (1983), 208–229.
- **[Ph]** F. Pham, *Formules de Picard–Lefschetz généralisées et ramification des intégrales*, Bull. Soc. Math. France **93** (1965), 333–367.
- **[Ran]** Z. Ran, *Cycles on Fermat hypersurfaces*, Compositio Math. **42** (1980/81), no. 1, 121–142.
- **[Shi79]** T. Shioda, *The Hodge conjecture for Fermat varieties*, Math. Ann. **245** (1979), no. 2, 175–184.

- **[Shi87]** T. Shioda, *Some observations on Jacobi sums*, in: Galois Representations and Arithmetic Algebraic Geometry, Adv. Stud. Pure Math. **12** (1987), 119–135. (Questions 7.2 and 7.4, p. 133, quoted as in [WDL].)
- **[SSvL]** M. Schütt, T. Shioda, R. van Luijk, *Lines on Fermat surfaces*, J. Number Theory **130** (2010), no. 9, 1939–1963. doi:10.1016/j.jnt.2010.01.008; arXiv:0812.2377. (Quoted as in [WDL].)
- **[SK]** T. Shioda, T. Katsura, *On Fermat varieties*, Tôhoku Math. J. (2) **31** (1979), no. 1, 97–115.
- **[WDL]** R. Amichis Luengo, *The Watermark and Double Ladder Theorems (Verified in Lean)*, Zenodo (2026). doi:10.5281/zenodo.23062529.